

Fine-mapping under a sparse genetic architecture

Iteration 004 benchmark — complete analysis of the sparse arm

fmbenchmark

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Abstract

This report analyses the *sparse* arm of the Iteration 004 fine-mapping benchmark: 9 design rows, 11,250 method fits per method, 18 methods, all fitted with in-sample LD. It covers overall accuracy, the conditions under which each method performs well, false-discovery and calibration behaviour, credible-set economics, and the three “senses” of variable importance (variance-explaining, decision-relevant, diagnostic). Every quantity derived from posterior inclusion probabilities is defined explicitly in §2 before it is used, including the estimator’s tie handling, its aggregation level, and the noise correction applied to the variance decomposition. The headline results are: (i) the number of causal variants S dominates the variance budget everywhere; (ii) the *presence* of annotations matters and their *strength* barely does; (iii) the Functional BEATRICE ablation is monotone and significant, with region sharing contributing more than annotations alone under continuous annotations; and (iv) the BEATRICE family buys ranking accuracy at a measurable cost in calibration.

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1 What was simulated, and what a “cell” is

The sparse arm holds the genetic architecture fixed at a *sparse* model: each region contains exactly S causal variants and no infinitesimal background. Nine design rows cross the annotation condition with the enrichment level:

rows	annotation type	enrichment folds
1	none	—
2–5	binary	2.7, 5.4, 8.1, 10.8
6–9	continuous	2.7, 5.4, 8.1, 10.8

Within every row the same sweep is run: $S \in \{1, 2, 3, 5, 10\}$, $\phi \in \{0.05, 0.1, 0.2, 0.4, 0.6\}$ (region heritability), 10 iterations, and 10 regions whose sizes are $\{500, 500, 750, 750, 1000, 1000, 1500, 1500, 2000, 2000\}$ variants, with $n = 5000$ individuals. That is $5 \times 5 \times 10 = 250$ scenarios per row and $250 \times 10 = 2,500$ fits per method per row.

The unit of analysis. A cell is a (row, S , ϕ , region size) combination. Each cell aggregates 10 iterations $\times$ 2 regions of that size = 20 fits. There are $9 \times 5 \times 5 \times 5 = 1,125$ cells per method. Cell means are the input to every ANOVA in §§8–10; individual fits are the input to the calibration counts in §6. The distinction matters and is stated wherever a number is quoted.

Two regions per size class. Region size is deliberately duplicated. The two regions of a given size share a size class but are *different draws* from the reference panel, which is what makes the

between-region variance component σ_u^2 identifiable (§2.5). Without it the split-plot correction could not be computed at all.

Stratification is not optional. Results are never pooled across annotation type. A method that cannot use annotations is not comparable to one that can, and the “none” arm additionally serves as a negative control: any method whose annotation machinery is inert there must reduce *exactly* to its annotation-blind counterpart. Two such identities are checked mechanically (§11).

2 From posterior inclusion probabilities to numbers

Every method returns, for each region, a vector $\text{PIP}_1, \dots, \text{PIP}_p \in [0, 1]$ — one posterior inclusion probability per variant — together with the truth $y_j \in \{0, 1\}$ known from the simulation. All metrics below are functions of that pair. This section defines each one and justifies the estimator, because several of the obvious implementations are wrong in ways that silently change conclusions.

2.1 Average precision (the ranking metric)

Average precision summarises the whole precision–recall curve without committing to a threshold, which matters here because methods’ PIP scales differ by orders of magnitude: any fixed-threshold comparison partly measures calibration rather than ranking.

Sort the variants by PIP descending. Let k index the distinct PIP values (so that all variants sharing a value form one block), tp_k the cumulative number of causals at the end of block k , n_k the cumulative number of variants, $\text{prec}_k = \text{tp}_k/n_k$ and $\text{rec}_k = \text{tp}_k/S$. Then

$$\text{AP} = \sum_k (\text{rec}_k - \text{rec}_{k-1}) \text{prec}_k, \quad \text{rec}_0 = 0. \quad (1)$$

Three implementation points are load-bearing.

It is a step integral, not a trapezoid. Equation (1) evaluates precision at the operating point reached, matching `sklearn.metrics.average_precision_score`. A trapezoidal rule interpolates between operating points that the classifier never actually occupies and is optimistically biased.

Ties are resolved by block, not by index. Several methods emit many exactly-tied PIPs. Implementing AP as “sort, then read off precision at the causal’s index” lets the sort’s tie-break — here genomic position — decide the score. Because causal variants are drawn uniformly at random this would not introduce a constant bias but would inject pure noise into precisely the comparisons the analysis exists to make. Collapsing tied variants into one block gives every member the same operating point, which is the only tie-handling that is invariant to input order.

AP is computed per fit and then averaged, never on a merged ranking. Pooling all variants from all regions into one ranking and computing a single AP is a different and incorrect quantity: it rewards a method for getting the *relative calibration across regions* right, which is not what region-level fine-mapping is asked to do. The two disagree materially. For two fits with $\text{AP} = 1$ and $\text{AP} = 1/3$, the correct answer is the mean $2/3 \approx 0.667$, whereas the merged ranking of the same data gives 0.700. Both values are checked against hand computation in the test suite.

A fit with no causal variants returns NA rather than 0, and NAs are excluded from means with the count reported alongside, so a method is never credited or penalised for a fit that posed no question.

2.2 Thresholded discovery: FDR and its honest companion

For a threshold t let $\mathcal{S}_t = \{j : \text{PIP}_j \geq t\}$, $n_t = |\mathcal{S}_t|$ and $\text{tp}_t = \sum_{j \in \mathcal{S}_t} y_j$. Then

$$\text{FDR}_t = \frac{n_t - \text{tp}_t}{n_t}, \quad \text{reported jointly with } n_t. \quad (2)$$

Reporting FDR alone is not meaningful: a method that never crosses t has $\text{FDR}_t = 0$ by convention and $n_t = 0$, and would otherwise win the comparison by abstaining. The pair (FDR_t, n_t) is therefore always presented together, and n_t appears in the credible-set analysis (§2.4) as the follow-up burden.

We also compute the **honesty ratio**. A Bayesian method that emits PIP values is making a falsifiable claim about its own error rate: the expected number of false positives among $\mathcal{S}_t$ is $\widehat{\text{FP}}_t = \sum_{j \in \mathcal{S}_t} (1 - \text{PIP}_j)$. Comparing that to the realised count gives

$$\rho_t = \frac{\text{FP}_t}{\widehat{\text{FP}}_t}, \quad (3)$$

so $\rho_t = 1$ means the method’s own uncertainty statement was accurate, $\rho_t > 1$ means it was overconfident by that factor. This is stricter than FDR: it holds the method to *its own* claim rather than to an external threshold.

2.3 Calibration

Two complementary summaries are used, because each is separately gameable.

Total mass ratio. $\sum_j \text{PIP}_j / S$. A coherent posterior over which variants are causal should place total mass equal to the expected number of causals. A ratio of 2 means the method claims twice as many causal variants as exist.

Banded reliability. Variants are binned by PIP into $[0, 0.1)$, $[0.1, 0.5)$, $[0.5, 0.9)$, $[0.9, 1]$. Within a bin we accumulate the count n_b , the number truly causal c_b , and $\sum \text{PIP}$. Calibration compares the realised rate c_b/n_b to the claimed mean $\sum \text{PIP}/n_b$; this is the reliability diagram in Figure 7. We report ECE_{hi} , the expected calibration error restricted to bins with $\text{PIP} \geq 0.1$, because the $[0, 0.1)$ bin contains the overwhelming majority of variants and an unrestricted ECE is dominated by the trivially-correct statement that almost nothing is causal.

Why both, and why RES. Consider a predictor emitting $\text{PIP}_j = S/p$ for every variant. Its mass ratio is exactly 1 and its calibration error is exactly 0 — it is perfectly calibrated and completely useless. The Murphy decomposition of the Brier score separates this out:

$$\text{BS} = \underbrace{\text{REL}}_{\text{miscalibration}} - \underbrace{\text{RES}}_{\text{resolution}} + \underbrace{\text{UNC}}_{\text{base rate}}, \quad (4)$$

where RES measures how far the method’s conditional rates depart from the base rate. The uniform predictor has $\text{REL} = 0$ and $\text{RES} = 0$. Resolution is therefore the component that detects calibration achieved by saying nothing, and it is reported in Table 3 and Figure 9 alongside REL.

2.4 Credible sets

For a target mass α , variants are added in decreasing PIP order until the cumulative mass first reaches α ; the resulting set size is $|CS_\alpha|$ and coverage is the indicator that the set contains a causal variant, averaged over fits. When a posterior’s *total* mass never reaches α the set is the whole region and a flag `set_reached` records it — this is a result, not a missing value, and suppressing it would let a maximally diffuse posterior masquerade as a confident small set. Size and coverage are always read together (Figure 10): a set can be made to cover by being large, and small by abandoning coverage.

2.5 The replication structure and the noise floor

The design is a split plot. Two variance components matter: σ_ε^2 , the between-iteration variance at fixed region, and σ_u^2 , the between-region-draw variance. A factor whose levels are compared *within* the same region draws (namely S and ϕ , which are re-simulated per scenario) sees only σ_ε^2 . A factor whose levels necessarily use *different* region draws — region size, and any term containing it — additionally carries σ_u^2 . Those are “whole-plot” terms.

For a term T with df_T degrees of freedom, the expected sum of squares under the null is $\text{df}_T \lambda_T$ with

$$\lambda_T = \frac{\sigma_\varepsilon^2}{20} + \mathbf{1}\{T \text{ whole-plot}\} \cdot \frac{m \sigma_u^2}{2}, \quad m = 25, \quad (5)$$

the 20 being the fits per cell and the 2 the regions per size class. The noise-corrected share is

$$\omega_T^2 = \frac{\text{SS}_T - \text{df}_T \lambda_T}{\text{SS}_{\text{tot}}}. \quad (6)$$

Why the correction is necessary. Without it, a factor that explains nothing still receives a share proportional to its degrees of freedom, and whole-plot terms receive a large one. In simulation, an uncorrected null factor recorded a 14.6% share where the truth was zero; corrected, it averaged -0.6% over replicates. Negative values are reported as computed and never clamped: a systematically negative share is a diagnostic that $\hat{\lambda}$ is too large, and hiding it would hide the diagnostic.

Estimating σ_u^2 requires a bias correction. The naive pooled estimator over J region draws in K groups is biased low by the factor $c = K(J - 1)/(JK - 1) = 0.758$ at $J = 4$ — because the pooled mean absorbs part of the between-draw signal. Dividing by c restores unbiasedness; verified over 120 replicates, recovering 0.0438 against a true 0.0400 where the naive estimator gave 0.0332.

2.6 What each “sense” asks

Three different questions are routinely conflated under “which variables matter”. They are kept separate throughout.

Sense V — variance-explaining. Of the variation in a metric across cells, how much does each design factor account for, after the noise correction of (6)? This bounds every claim below it: a noise floor of 50% means no decomposition can explain more than half.

Sense D — decision-relevant. Does changing a factor change *which method you would choose*? Measured as the rate at which the best-performing method flips between adjacent factor levels, split into a raw rate and a gated rate that counts only flips whose margin survives a cross-fitted paired test at $z = 2$. The gap between them is the quantity of interest: it is how much of the apparent structure is noise.

Sense G — diagnostic. For a specific method, which factors hurt *it* relative to what was achievable? Measured by $\kappa = (\text{method} - \text{baseline})/(\text{ceiling} - \text{baseline})$, the fraction of available headroom captured, with SuSiE as baseline and PolyFun-oracle as ceiling.

Sense V and Sense D can disagree entirely, and in this dataset they do. A factor can move a metric a great deal while never changing the winner (every method degrades together), and a factor with a small main effect can flip the winner if it acts differently on different methods. Reporting only one of them overstates what is known.

3 Completeness: what is present and what is absent

Sixteen of the eighteen methods have all 1,125 cells populated with fit-failure rates between 0.0% and 0.5%. Two do not, and in both cases the absence is structural rather than a data defect:

- **Funmap** requires functional annotations and cannot run at all on the “none” arm (0/125 cells). It is complete (500/500) on both annotated arms. Its row on the unannotated stratum is *empty*, not *poor*, and no performance claim may be inferred from it.
- **FINEMAP-inf** is missing exactly 45 cells, all at $\phi = 0.6$ and $S = 1$ — nine rows $\times$ five region sizes. It returns an all-NA posterior without raising an error when a single variant carries 60% of regional heritability. This is recorded through a dedicated `bad_pip` flag rather than being averaged away, so the loss is countable per method and per factor level.

CARMA was dropped from the method set mid-iteration on measured cost and is excluded from every analysis here; the eighteen methods reported are the complete set actually fitted.

4 Overall accuracy

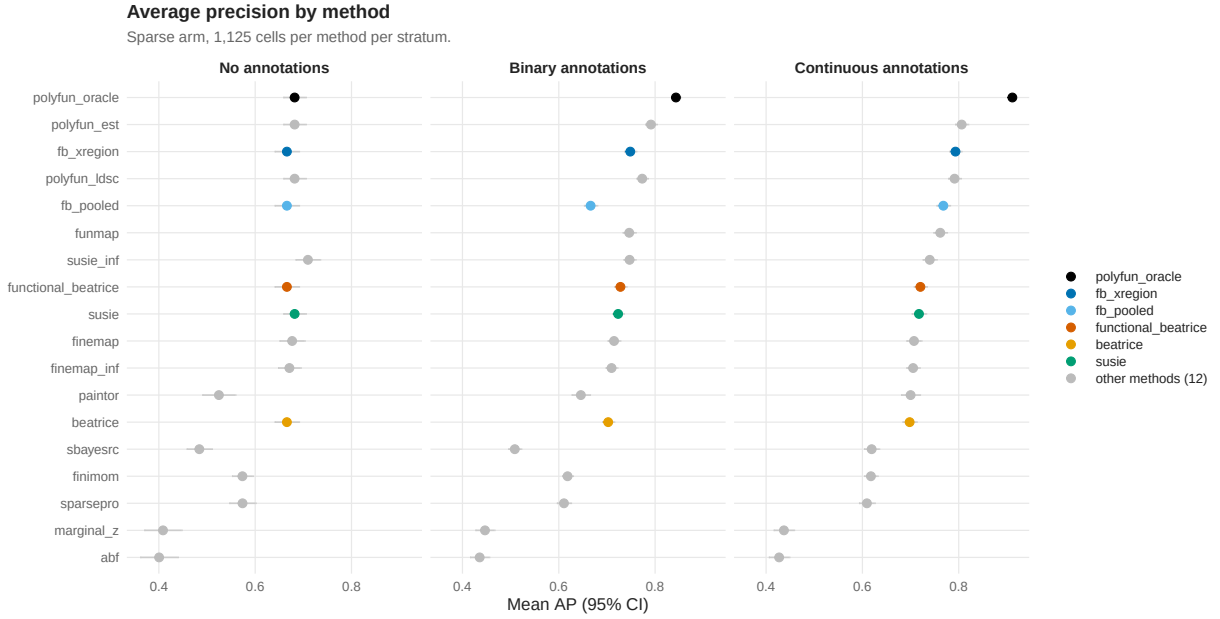


Figure 1: Mean average precision by method, with 95% intervals over the 1,125 cells of each stratum. PolyFun-oracle reads the simulated annotation weights and is a ceiling, not a deployable competitor.

Three features of Figure 1 are worth stating plainly.

The oracle gap is real but not enormous. PolyFun-oracle sits above every deployable method on the annotated strata, but the best estimated-prior methods recover a substantial share of that headroom. On the unannotated arm the oracle has no information to exploit and collapses onto SuSiE exactly — a mechanical check, not a coincidence (§11).

Annotation-aware methods separate from annotation-blind ones only when annotations exist. The ordering on the “none” panel is essentially a ranking of sparse-regression quality; the ordering on the annotated panels reorders substantially.

The **marginal baselines are informative**. `marginal.z` and `abf` sit far below everything else, which confirms the design poses a genuine fine-mapping problem: LD is doing real work, and simply ranking by marginal significance is not competitive.

5 When does each method do well?

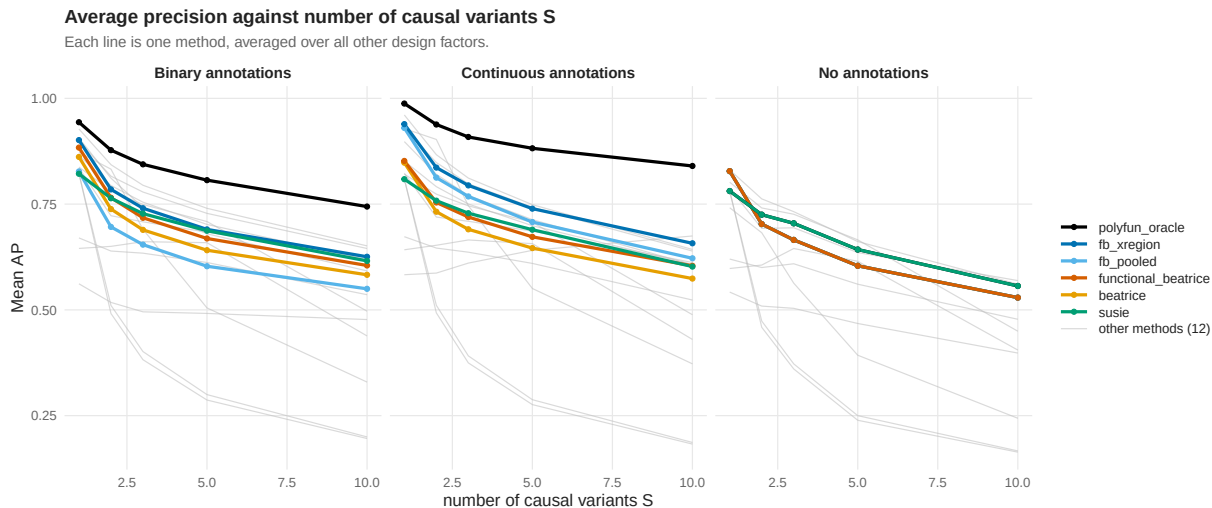


Figure 2: AP against the number of causal variants. Every method degrades as S grows; the spread between methods also widens.

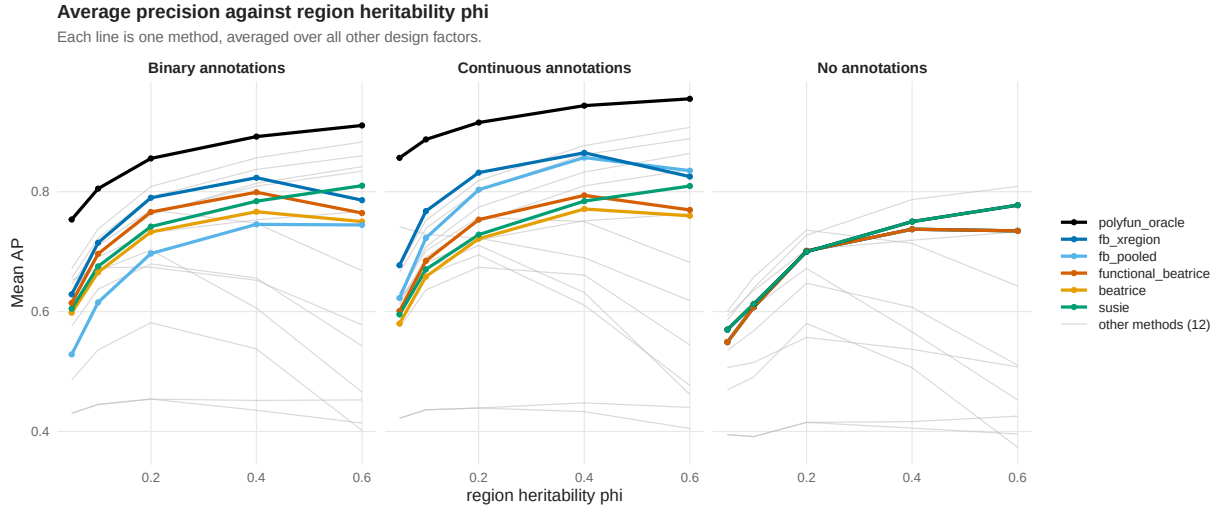


Figure 3: AP against region heritability. More signal helps everything, with diminishing returns at the top of the range.

The practical summary is that **sparsity is the binding constraint**. Moving from $S = 1$ to $S = 10$ costs far more accuracy than moving from $\phi = 0.6$ to $\phi = 0.05$, and both dwarf region size and enrichment strength. For a practitioner this reframes the question: the useful prior information is how many independent signals a locus is likely to contain, not how large the locus is.

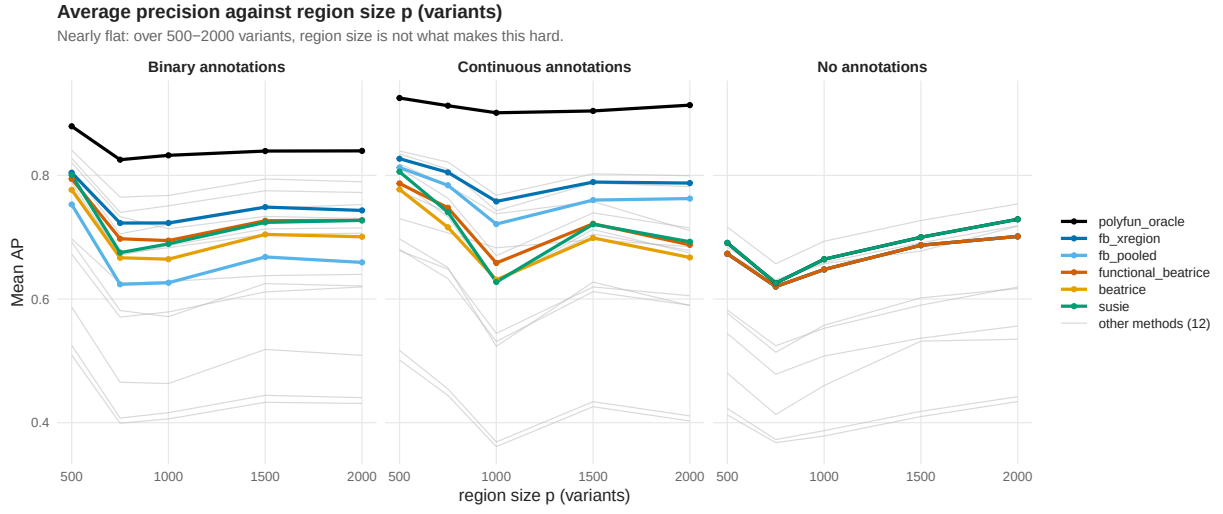


Figure 4: AP against region size. Nearly flat — over 500–2000 variants, region size is not what makes fine-mapping hard here.

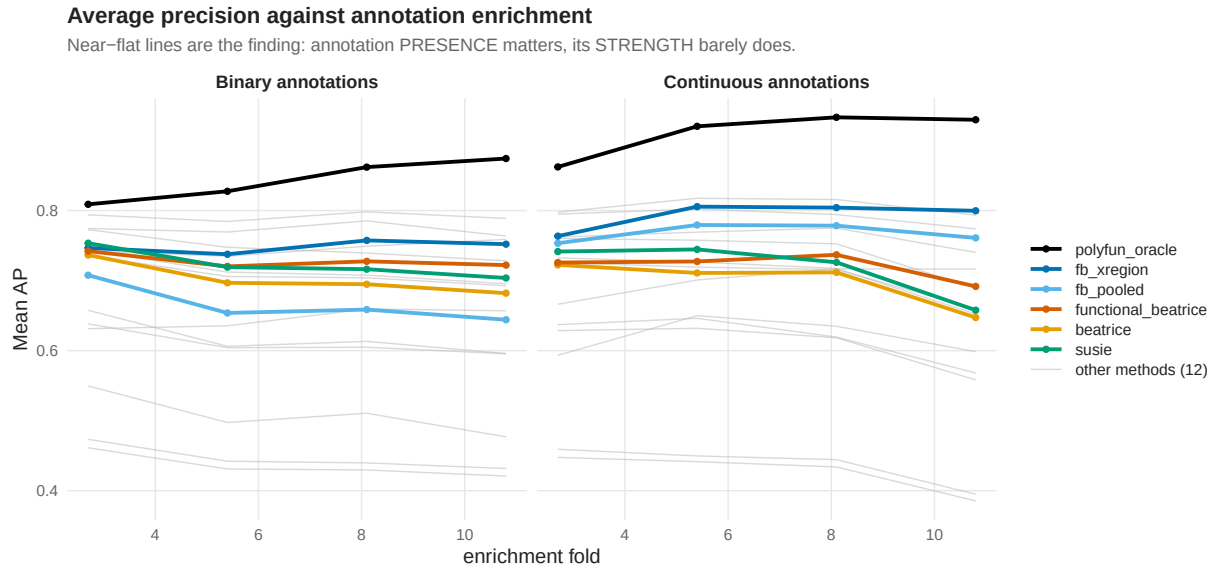


Figure 5: AP against annotation enrichment fold. The near-flat lines are the result: what matters is that annotations exist, not how strongly they are enriched over the range swept.

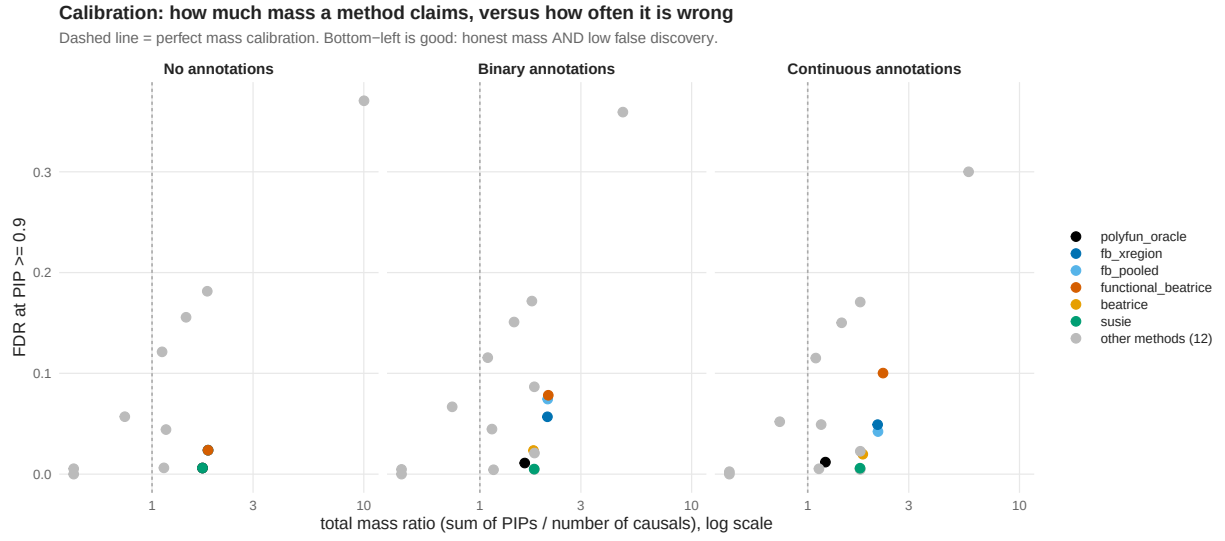


Figure 6: Mass claimed versus errors made. The dashed vertical line is perfect mass calibration; the bottom-left corner is the desirable region.

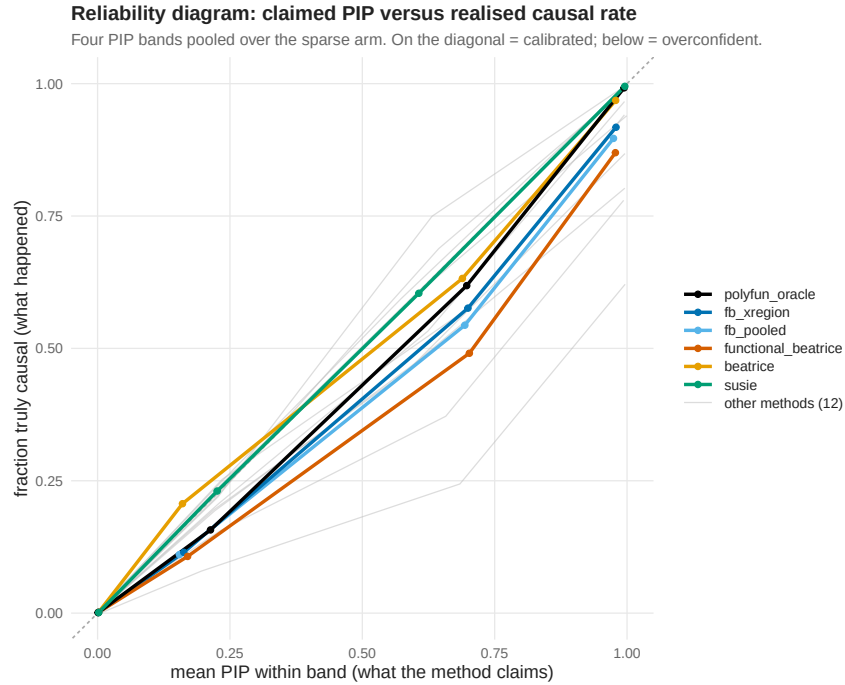


Figure 7: Reliability diagram over the whole sparse arm. Points below the diagonal are overconfident: the method claimed more probability than the realised causal rate justified.

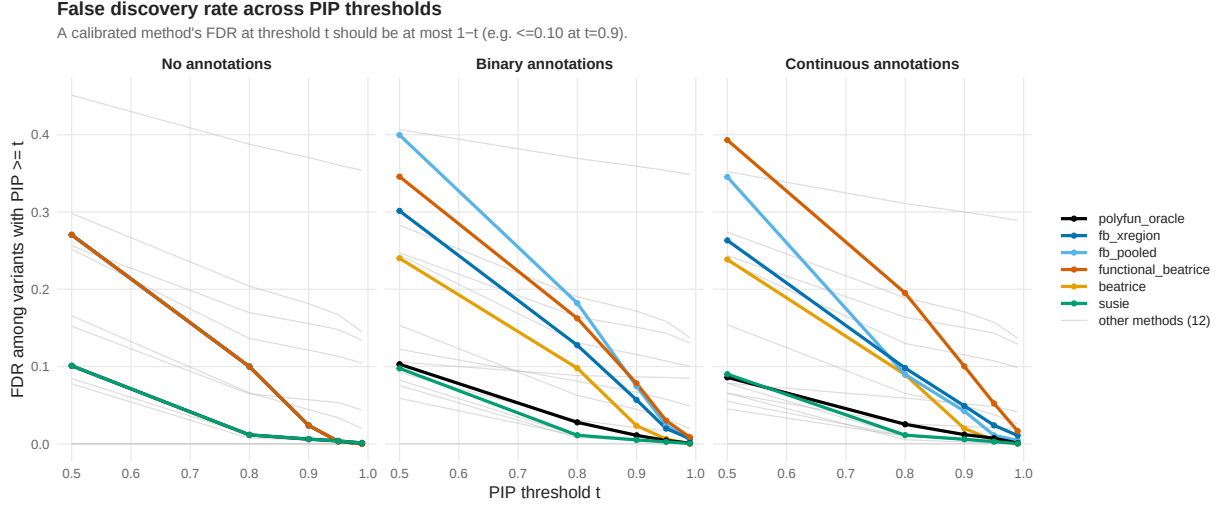


Figure 8: FDR as a function of the PIP threshold. A calibrated method should satisfy $FDR_t \leq 1 - t$.

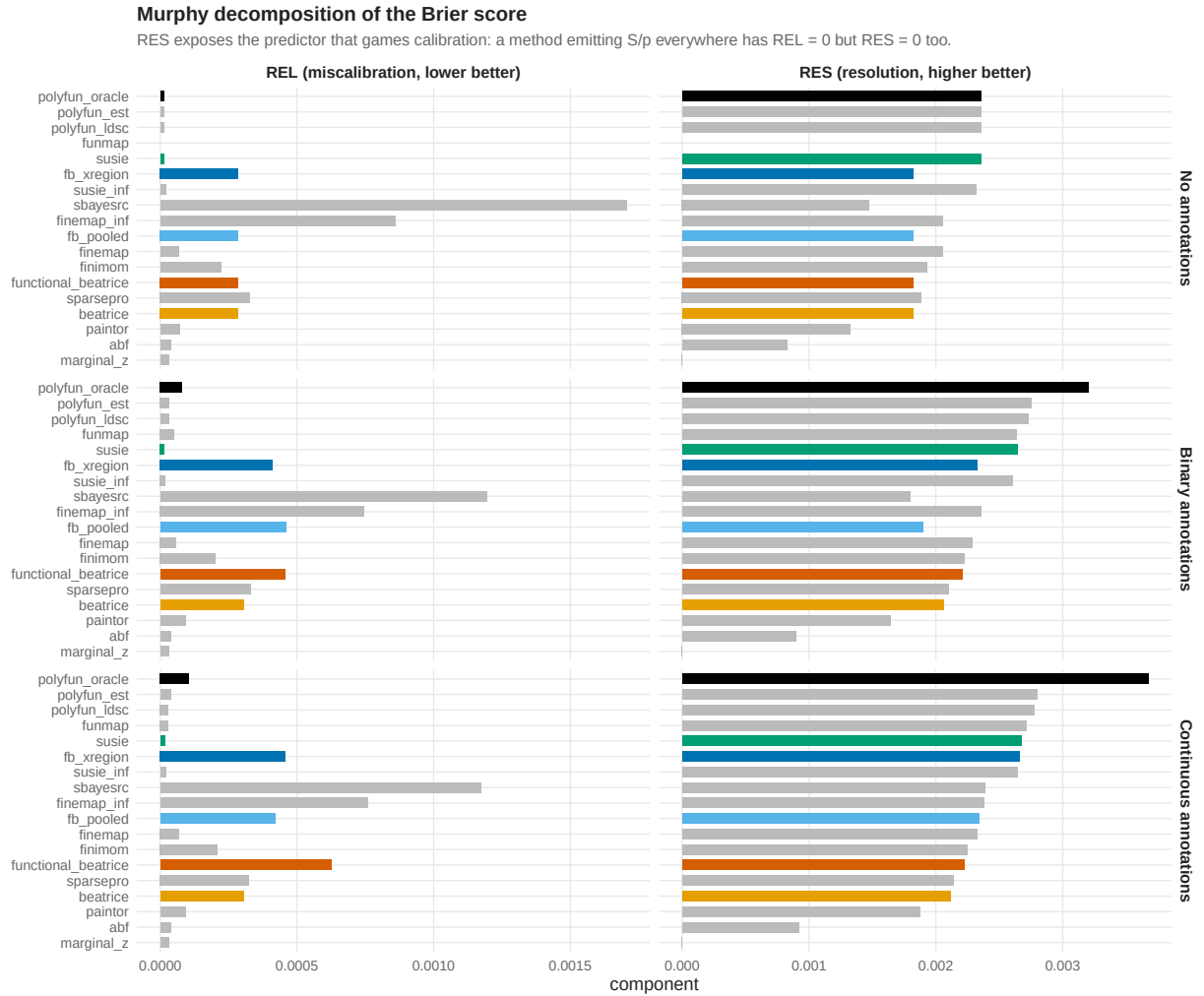


Figure 9: Murphy decomposition. REL is miscalibration (lower is better); RES is resolution (higher is better). RES is what separates a genuinely informative posterior from one that is calibrated by being vague.

6 False discovery and calibration

The central trade-off in Figure 6 is that **ranking accuracy and calibration are not the same axis and the leaders differ**. Methods can be grouped:

- **Conservative and well-calibrated:** SuSiE-inf and FINEMAP hold mass ratios near 1.1–1.2 with low FDR, at some cost in AP.
- **Accurate and mass-inflated:** the BEATRICE family achieves high AP while claiming roughly twice the mass that exists, with FDR at $\text{PIP} \geq 0.9$ an order of magnitude above the PolyFun family.
- **Diffuse:** sbayesrc spreads mass extremely widely (ratios in the multiples), which shows up as both poor calibration and a large credible-set burden.
- **Abstaining:** marginal_z and abf place very little mass anywhere; their apparent FDR advantage is a consequence of rarely crossing the threshold at all, which is exactly why n_t is reported alongside.

7 Credible sets: the follow-up burden

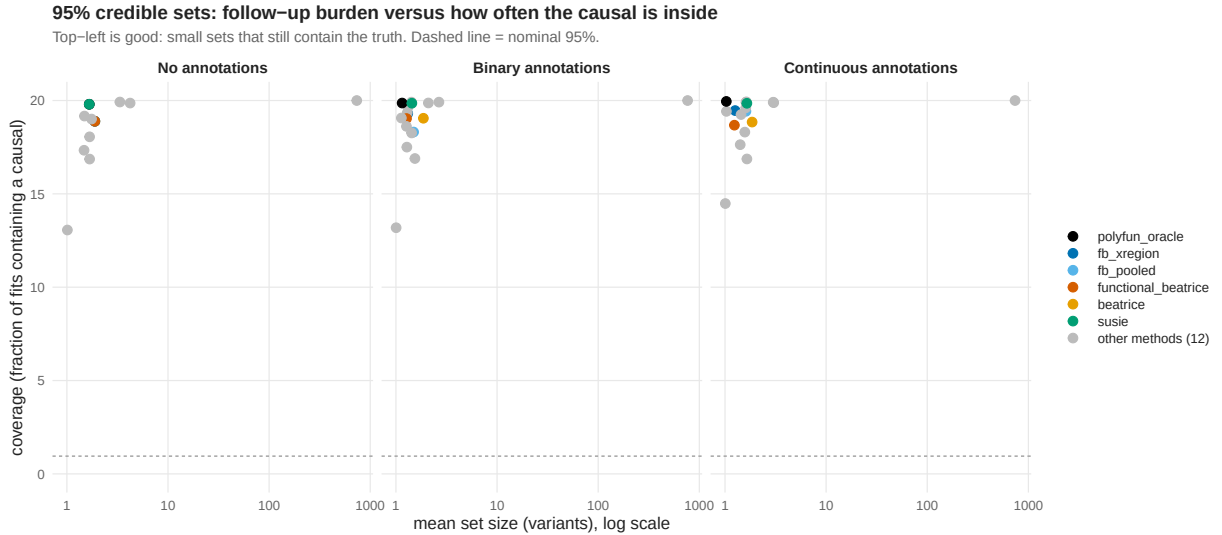


Figure 10: 95% credible sets. The practitioner’s question is the pair: how many variants must be taken to the bench, and how often is the causal one among them.

Coverage below the nominal 95% combined with small sets indicates a posterior that is too sharp; coverage at nominal with very large sets indicates one that is too diffuse. Neither number alone identifies which situation a method is in, which is why Figure 10 plots them jointly and on a log size axis.

8 Sense V: what moves the metric

8.1 How the decomposition is computed, in full

Every number in this section descends from the per-fit AP values of §2.1 by the following chain. Constants are named as they appear in the code: $N_ITER=10$, $N_REGIONS=2$ (regions per size class), $N_FITS=20$, $M_SHARED=25$, $SUBPLOT_FACTORS=\{S, \phi\}$.

Step 1 — form cell means. A cell is the tuple (job_dir, S , ϕ , region size). Its value is the mean of the N_FITS = 20 fits it contains: 10 iterations $\times$ 2 regions of that size class. This table (L3) is the only input to the ANOVA; individual fits are used solely to estimate the two variance components below.

Step 2 — estimate σ_ε^2 , the iteration variance. Group the fit table by

$$\text{KEY} = (\text{job_dir}, \text{region size}, \text{region_idx}, S, \phi),$$

which pins down *one physical region at one design point*, leaving only the iteration index free. Within each group take the sample variance across its 10 iterations; $\hat{\sigma}_\varepsilon^2$ is the mean of those variances. Note what is held fixed: the region, its LD, its variants and its annotations are literally identical across those 10 values, so this is pure replicate error from the phenotype draw.

Step 3 — estimate σ_u^2 , the between-region-draw variance. This cannot be read off directly, because a region draw is never replicated within a size class — there are exactly two, and they differ. It is recovered from their contrast:

1. Average over iterations within each KEY group, giving a region mean $\bar{y}(\text{job}, \text{size}, \text{idx}, S, \phi)$.
2. Reshape wide on `region_idx` and difference the pair: $d = \bar{y}_1 - \bar{y}_2$ for each (job, size, S, ϕ).
3. In expectation $\text{Var}(d) = 2\sigma_u^2 + 2\sigma_\varepsilon^2/\text{N_ITER}$, so subtracting the second term and halving would give σ_u^2 — except for the bias in Step 4.
4. Do this separately for each region size, then average the five estimates into $\bar{\sigma}_u^2$.

Step 4 — debias the pair variance. The two region draws of a size class are *reused* across all K combinations of (S, ϕ) within a job, so those K differences share a common draw-level offset D_j and are not independent. With J job rows and K cells per job the naive $\text{Var}(d)$ is biased low by

$$c = \frac{K(J-1)}{JK-1}, \quad \hat{\sigma}_D^2 = \max\left(0, \frac{\text{Var}(d) - 2\sigma_\varepsilon^2/\text{N_ITER}}{c}\right), \quad \hat{\sigma}_u^2 = \hat{\sigma}_D^2/2.$$

At $J = 4$ this is $c = 0.758$, i.e. the naive estimator recovers only about three-quarters of the true component. Verified over 120 simulated replicates: corrected 0.0438 against a true 0.0400, where the naive estimator gave 0.0332.

Step 5 — fit the saturated ANOVA on cell means. `aov(response ~ f1 * f2 * ...)` over the stratum's free factors, all interactions included. The design is complete and balanced, so the model is saturated: **there is deliberately no residual line**, every degree of freedom belongs to a named term, and the orthogonal sums of squares partition SS_{tot} exactly. If a **Residuals** row does appear the code warns and drops it, because its presence means the stratum is not balanced and the closed forms below no longer hold.

Step 6 — classify each term.

$$T \text{ is whole-plot} \iff T \text{ contains no factor from } \{S, \phi\}.$$

The logic runs through what a contrast differences away. S and ϕ are re-simulated per scenario *on the same region draw*, so any contrast involving them cancels the draw exactly and sees only iteration noise. A term built solely from region size and enrichment compares values computed on *different* region draws and therefore carries σ_u^2 as well. Note the consequence: **enrich** alone and **enrich:region_size** are whole-plot, whereas **S:region_size** is not, despite containing region size.

Step 7 — the noise expectation and the corrected share.

$$\lambda_T = \frac{\sigma_\epsilon^2}{N_FITS} + \mathbf{1}\{T \text{ whole-plot}\} \cdot \frac{M_SHARED \cdot \bar{\sigma}_u^2}{2}, \quad \omega_T^2 = \frac{SS_T - df_T \lambda_T}{SS_{tot}}.$$

$N_FITS=20$ is the number of fits behind each cell mean; $M_SHARED=25=5 \times 5$ is the number of (S, ϕ) cells sharing one region draw, and the $/2$ is the two regions per size class. The noise floor is $\sum_T df_T \lambda_T / SS_{tot}$, and it is reported split into its iteration and region parts.

Why cell means rather than one replicate-level ANOVA. Both are defensible in principle; only one is safe here. At replicate level each noise source enters a mean square multiplied by the number of rows sharing that draw — hundreds of them — so subtracting a cell-mean-scale $\sigma_u^2/2$ under-corrects by orders of magnitude *while still looking like a correction has been applied*. Working on cell means keeps every quantity on one scale, which is why the closed form above is exact rather than approximate.

Sobol indices. $V_T = \max(\omega_T^2, 0)$ — this is the *only* place a clamp is applied, and it is applied because the indices are ratios that would otherwise be uninterpretable. Then $S_i = V_i / \sum_T V_T$ and $S_{Ti} = \sum_{T \ni i} V_T / \sum_T V_T$, so $S_{Ti} - S_i$ is the share of variance factor i participates in only through interactions. Everywhere else in this report negative ω^2 is reported as computed.

8.2 The variance budget

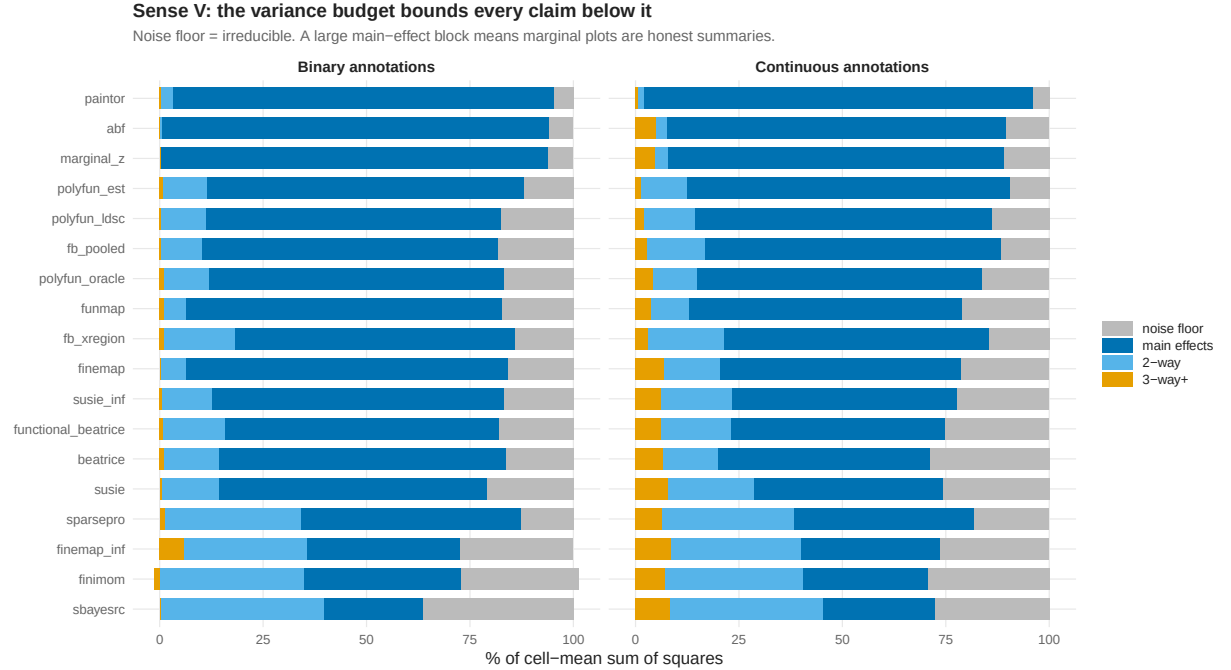


Figure 11: The closed variance budget for AP. Everything below the noise floor is irreducible at this replication level.

Averaged over methods:

stratum	noise floor	main effects	2-way	3-way+
binary annotations	17.0%	67.6%	14.7%	0.8%
continuous annotations	18.8%	59.6%	16.5%	5.1%

Two consequences. First, roughly **a sixth of the cell-to-cell variation is irreducible noise** — no factor decomposition can account for more than about 82%, and any method whose “explained” shares approach 100% is overfitting the noise. Second, **main effects carry the clear majority** (60–68%) against 15–17% for all two-way terms and under 5% for everything higher. That is the licence for reading the marginal plots of §5 at face value: the conditional structure exists but is roughly a quarter the size of the marginal structure.

8.3 Different metrics have different drivers

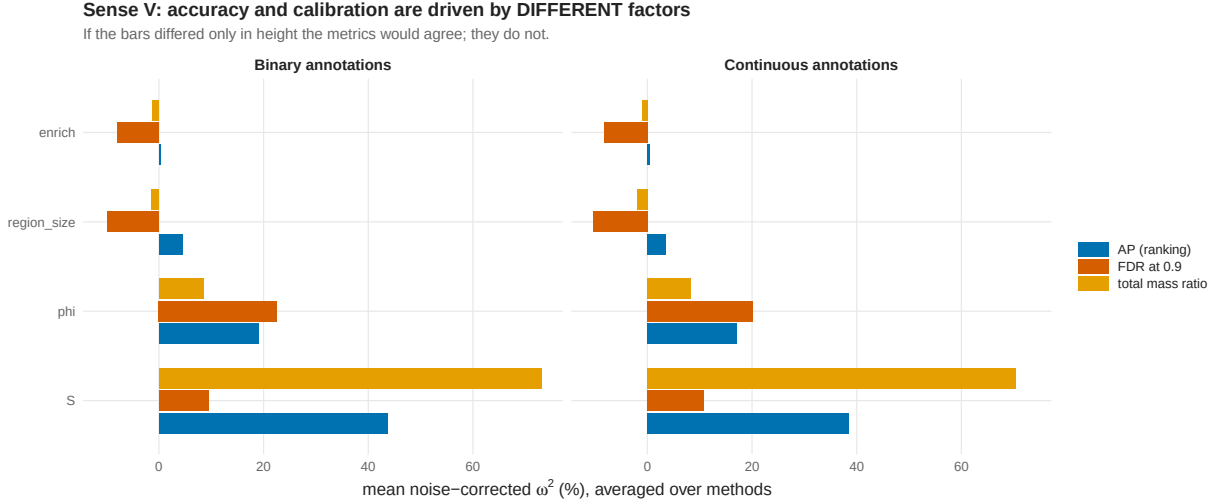


Figure 12: Noise-corrected ω^2 for three responses. The factor ordering changes with the response, so “what matters” is metric-dependent.

This is the result that a single-metric analysis would have missed. Pooling the two annotated strata, the mean ω^2 (%) is:

factor	AP (ranking)	FDR at 0.9	total mass ratio
S	41.1	10.1	71.8
ϕ	18.0	21.3	8.4
region size	4.0	-10.1	-1.7
enrichment	0.4	-8.0	-1.1

Ranking accuracy and mass calibration are both governed by S (41% and 72%), which is intuitive: how much total probability a method should emit is essentially a statement about how many causals there are. **False discovery is governed instead by ϕ** (21%, with S at only 10%). Signal strength, not signal count, is what determines whether a method’s confident calls are right. A benchmark reporting AP alone would conclude that ϕ is a second-order factor; for the metric a practitioner actually acts on — the error rate among high-PIP calls — it is the leading one.

The negative shares are a diagnostic, and are reported unclamped. Region size and enrichment take clearly negative ω^2 for FDR (−10.1 and −8.0). By construction that means $\text{df}_T \lambda_T$ exceeded SS_T , i.e. the estimated noise for those terms is larger than the variation observed. FDR at a fixed threshold is a ratio of small counts and is far noisier per cell than AP, so $\hat{\sigma}_\epsilon^2$ is large relative to the between-cell structure. The honest reading is that **the FDR decomposition is not reliable at this replication level**, and we report the negative numbers rather than clamping them precisely so that this is visible.

8.4 How conditional is each effect?

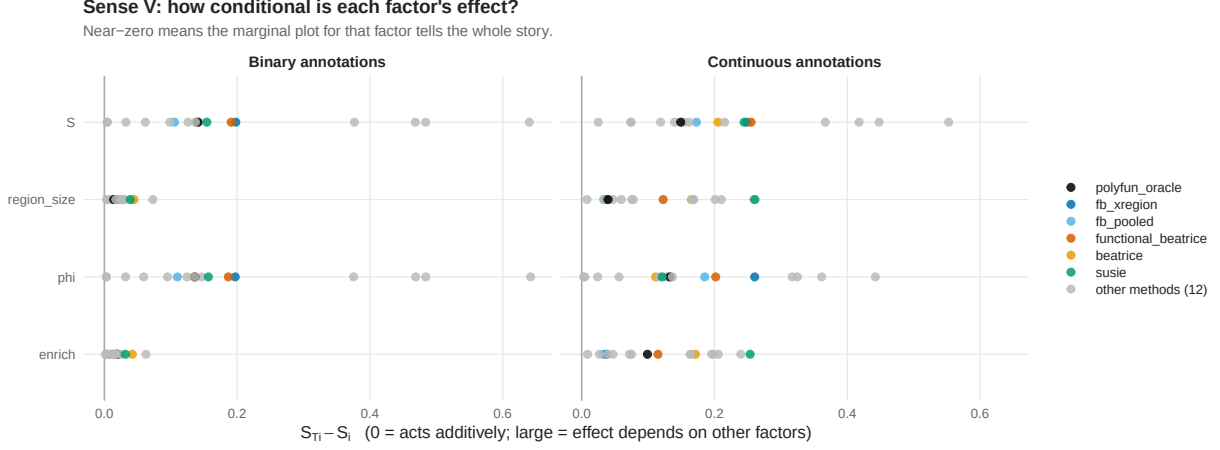


Figure 13: $S_{Ti} - S_i$ per factor and method: the share of variance a factor is involved in through interactions, over and above its own main effect.

S_i is the factor's own main-effect share; S_{Ti} is its total share including every interaction it appears in. The difference is how much of the factor's influence is conditional on the others. Values near zero mean the marginal plot is the whole story; large values mean a claim about that factor must be stated conditionally. The dominant interaction throughout is $S:\phi$, consistent with per-causal heritability ϕ/S being closer to the quantity that actually governs difficulty than either factor separately.

8.5 Caveats specific to this section

- **The unannotated stratum has no noise floor.** It contains a single design row, so there is only one group from which to estimate σ_u^2 and the between-region-draw component is unidentifiable. Every budget and corrected share in this section is therefore computed on the binary and continuous strata only. Uncorrected shares for the **none** arm would be inflated, particularly for whole-plot terms, and are not reported.
- **Range dependence.** ω^2 is relative to the input distribution, uniform over the levels swept. S spans tenfold, enrichment fourfold, so part of S 's dominance is range. Always “over the range swept”.
- **Metric scale.** ω^2 is not invariant to monotone transforms, and bounded metrics saturate. A factor operating where AP is already near its ceiling looks unimportant because there is no room left to move.

9 Sense D: what changes the decision

9.1 How a cell's winner is decided, in full

The unit is again the cell (job_dir, S , ϕ , region size), but here it is *not* collapsed to a mean. The procedure per cell is:

Step 1 — build the replicate matrix. Reshape to a matrix Y with one row per iteration ($R = 10$) and one column per method, entry $Y_{im} = \text{AP of method } m \text{ on iteration } i$. Because all methods ran on the same simulated data, row i is a genuine paired observation across methods.

Columns that are entirely non-finite are dropped; fewer than two surviving columns means the cell is skipped.

Step 2 — the nominal winner. $\text{winner} = \arg \max_m \bar{Y}_{.m}$, the column mean over all 10 iterations. This is the quantity the regime maps display, and on its own it is **not** evidence of anything — the maximum of M noisy means is upward biased by construction.

Step 3 — cross-fitted testing. If $R < 4$ the cell is marked undecided immediately. Otherwise the 10 iterations are randomly permuted (once, under `set.seed(20260806)`, so the result is reproducible) and split into halves A and B . Then, in each direction:

1. Compute column means on the *selection* half.
2. Take the top two methods *by that half only*.
3. On the *other* half form the paired differences $d_i = Y_{i,\text{top1}} - Y_{i,\text{top2}}$.
4. Declare decided iff $|\bar{d}| > z \cdot \text{SE}(\bar{d})$ with $z = 2$ and $\text{SE} = \text{sd}(d)/\sqrt{n}$.

Step 4 — require agreement. The cell counts as decided only if **both** directions (A selects and B tests, then B selects and A tests) declare decided *and* both name the same winner. Any disagreement means the apparent winner depended on which half happened to be used for selection.

Why cross-fitting is necessary. Selecting the top two and testing them on the same data guarantees a positive margin: the selection has already sorted the noise. Splitting selection from testing removes that bias, and requiring both directions to agree removes the residual luck of a single split. This is why the gated rates are so much lower than the raw ones — the raw rate is measuring exactly the bias this procedure is designed to eliminate.

Step 5 — the reported margin. $\bar{d}$ ($\bar{d}$ bar) is computed on *all* 10 rows as $\overline{Y_{\text{winner}}} - \overline{Y_{\text{runner-up}}}$, using the cross-fitted runner-up. It is a descriptive effect size, not the test statistic.

Step 6 — flip rates. For a factor f : hold every other free factor fixed, walk adjacent levels of f , and record whether the winner changes. `flip_raw` is the fraction of such adjacent pairs whose nominal winner differs; `flip_gated` is the same fraction counting only pairs in which the relevant cells were decided. `pct_cells_decided` is reported separately as the unconditional decidability of the stratum.

9.2 Who actually wins

14 of the 18 methods win at least one cell, and only **53 of 1,125 cells (4.7%) have a statistically decided winner**.

By nominal win count the leaders are `polyfun_est` (340 cells), `paintor` (203) and `fb_xregion` (137). Among the 53 decided cells the ordering is different — `paintor` (20), `sbayesrc` (13), `polyfun_est` (10), `fb_xregion` (4).

That `sbayesrc` wins 13 decided cells despite ranking near the bottom on mean AP is worth stating plainly: a method can be poor on average and still be decisively best in a particular corner of the design. It is also a warning against reading the nominal win counts as a ranking — most of those 340 `polyfun_est` cells are ties dressed up as wins.

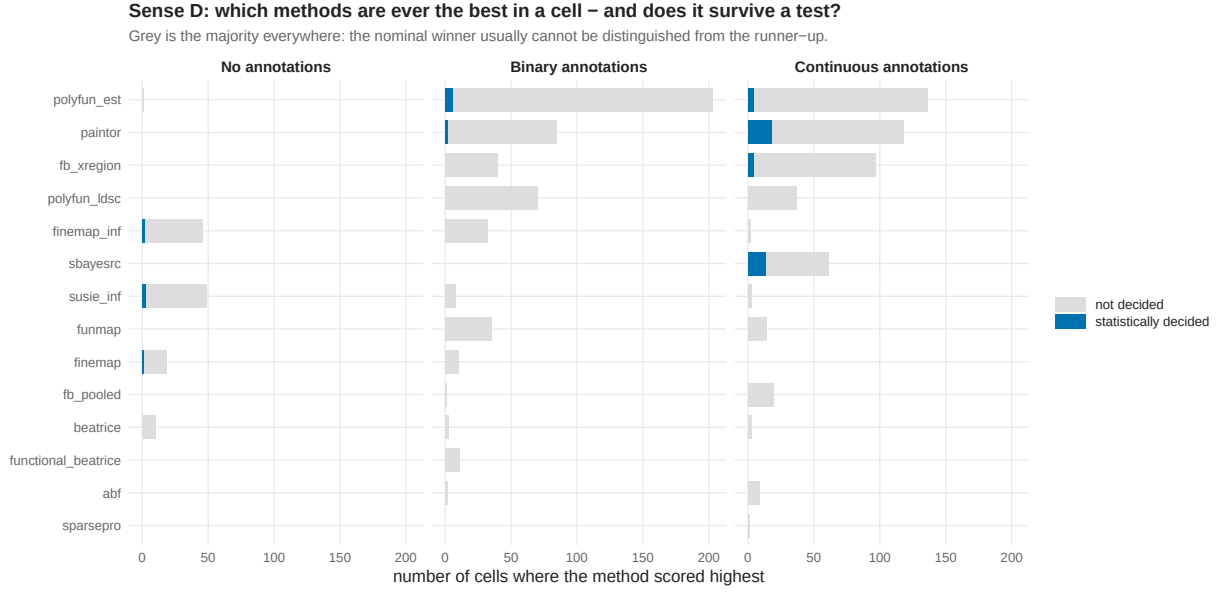


Figure 14: Cells won by each method, split by whether the win survives the paired test.

9.3 Why so little is decidable

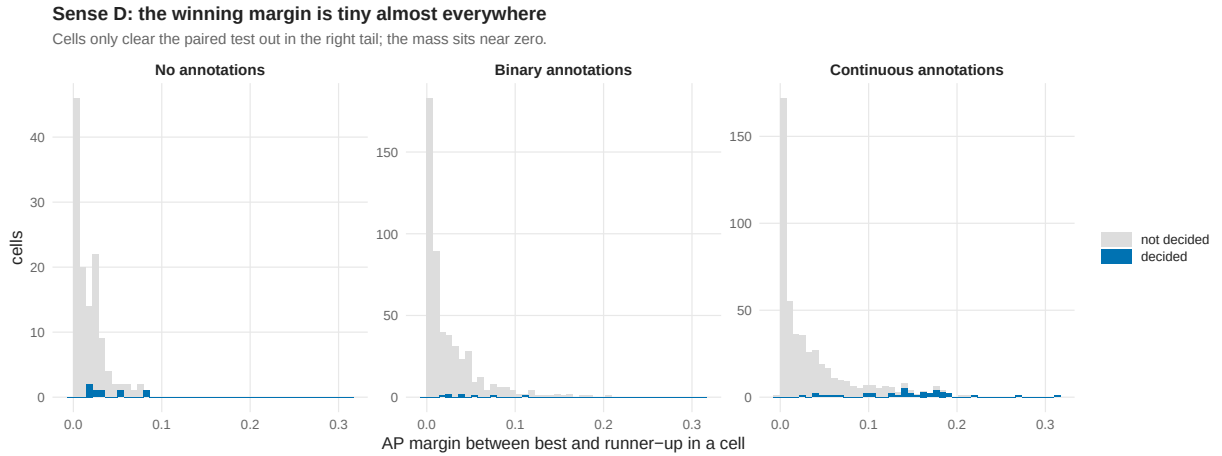


Figure 15: Distribution of the AP margin between the best and second-best method in each cell.

The mechanism is visible in the margins. The **median margin in undecided cells is 0.0128; in decided cells it is 0.1254** — a factor of ten. The mass of the distribution sits near zero, so the paired test is not being unreasonably strict; the methods genuinely are close together in most of the design.

What follows. At 20 fits per cell this benchmark **cannot resolve which of 18 methods is best within a single cell**, and regime maps built from modal winners must not be read as recommendations. This is a statement about power, not about the methods being equivalent: the paired two-method contrasts of §10 pool thousands of replicates and are decisive. Reporting only one of the two would misrepresent what the benchmark establishes.

10 Sense G: headroom and the Functional BEATRICE ablation

10.1 How κ and the controlled pairs are computed, in full

The input here is the *replicate-level* table (L2): one row per (cell, iteration, method), reshaped so that each (cell, iteration) carries one column per method. Because every method ran on the same simulated data, a row is a set of paired observations — same causal variants, same genotypes, same annotations, same phenotype draw.

Step 1 — headroom. Per row,

$$h = \text{AP}_{\text{polyfun_oracle}} - \text{AP}_{\text{susie}}.$$

`susie` is the annotation-blind backbone — what a method achieves *without* using annotations at all — and `polyfun_oracle` receives the simulator’s true prior weights, so it is what is achievable *with perfect* annotation information. Their difference is the room annotations create on that particular replicate.

Step 2 — the gate. Rows with $h \leq \text{HEADROOM_EPS} = 0.02$ (or non-finite h) are excluded, because κ divides by h and is unstable as $h \rightarrow 0$. The excluded fraction and the median headroom are reported rather than dropped silently (below); on the unannotated arm the gate removes *everything*, which is correct: the ceiling and the baseline are the same method there, so $h \equiv 0$ and κ is undefined by construction.

stratum	replicate rows	excluded by the gate	median headroom
no annotations	1,250	100.0%	0.000
binary	5,000	29.4%	0.084
continuous	5,000	21.8%	0.166

The median headroom is itself a result: continuous annotations create twice the exploitable room that binary ones do (0.166 against 0.084), which is the backdrop against which the ablation gains in §10.3 should be read.

Step 3 — κ . On surviving rows, for each focus method $m \in \{\text{functional_beatrice}, \text{fb_xregion}, \text{fb_pooled}\}$,

$$\kappa_m = \frac{\text{AP}_m - \text{AP}_{\text{susie}}}{h}.$$

$\kappa = 1$ means m captured all the available headroom; $\kappa = 0$ means it did no better than ignoring the annotations; $\kappa < 0$ means it did worse than the annotation-blind baseline on that replicate.

Step 4 — the controlled pairs, which are NOT gated. Two contrasts are formed on *all* rows, with no headroom gate, because they are differences rather than ratios and have no denominator to destabilise:

$$\Delta_{\text{annot}} = \text{AP}_{\text{functional_beatrice}} - \text{AP}_{\text{beatrice}}, \quad \Delta_{\text{joint}} = \text{AP}_{\text{fb_xregion}} - \text{AP}_{\text{functional_beatrice}}.$$

Each isolates exactly one mechanism, because the two methods being differenced are identical in every other respect: same inference network, same hyperparameters, same data, same seed. Δ_{annot} turns the annotation prior on; Δ_{joint} changes that prior from per-region to shared-across-regions.

Step 5 — inference on the pairs. For a group of n paired rows, $\bar{\Delta}$, $\text{sd}(\Delta)$, $\text{SE} = \text{sd}/\sqrt{n}$ and $t = \bar{\Delta}/\text{SE}$. The pairing is what supplies the power: the between-scenario variance that dominates marginal comparisons cancels exactly, leaving only the within-replicate difference. This is why §10 resolves effects of 0.02 AP decisively while §9 cannot resolve a 0.13 gap between two *different* methods in a single cell — n differs by three orders of magnitude and the nuisance variance has been removed.

Step 6 — conditional breakdowns. The tables in §10.3 are $\bar{\Delta}$ recomputed within levels of one design factor at a time, pooling the two annotated strata. 114 of the 22,500 Δ values are missing (0.5%), all at $S \in \{5, 10\}$, corresponding to the BEATRICE family’s small fit-failure rate; those rows are dropped pairwise.

10.2 The ablation

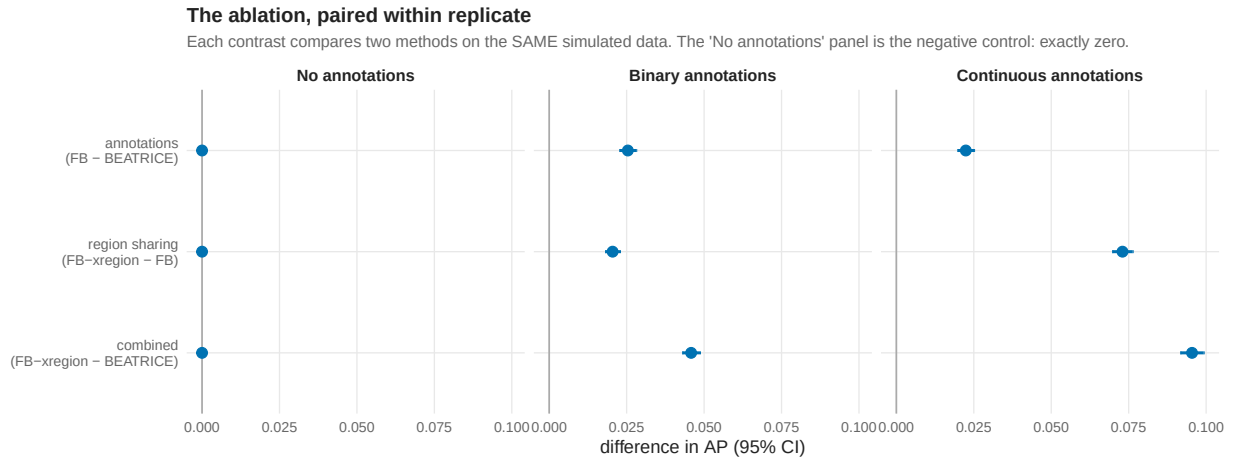


Figure 16: Paired contrasts with 95% intervals. The unannotated panel is the negative control.

Both stages contribute and the negative control is exact (Table 4): annotations add +0.025 AP under binary ($t = 18.4$) and +0.022 under continuous ($t = 16.4$); region sharing adds a further +0.021 ($t = 16.8$) and +0.073 ($t = 43.5$); combined, +0.046 and +0.095. On the unannotated arm all three contrasts are exactly 0.0000 ± 0.0000 over 1,250 paired replicates — the three variants are bit-identical there, so the measured gains cannot be an artefact of the extra machinery, of additional parameters, or of run-to-run variation.

10.3 Where the gains come from

Pooling the annotated strata, the conditional structure of the two mechanisms is different, and the difference is interpretable:

	$S = 1$	$S = 2$	$S = 3$	$S = 5$	$S = 10$
annotations	+0.0131	+0.0243	+0.0288	+0.0272	+0.0262
region sharing	+0.0523	+0.0511	+0.0487	+0.0441	+0.0370

Region sharing is worth most when regions contain fewest causals, and decays monotonically as S grows ($+0.052 \rightarrow +0.037$). That is the behaviour the mechanism predicts: a region with a single causal variant carries almost no internal information about which annotations matter, so borrowing the enrichment estimate from nine other regions is worth a great deal; a region with ten causals can estimate it locally and has less to gain.

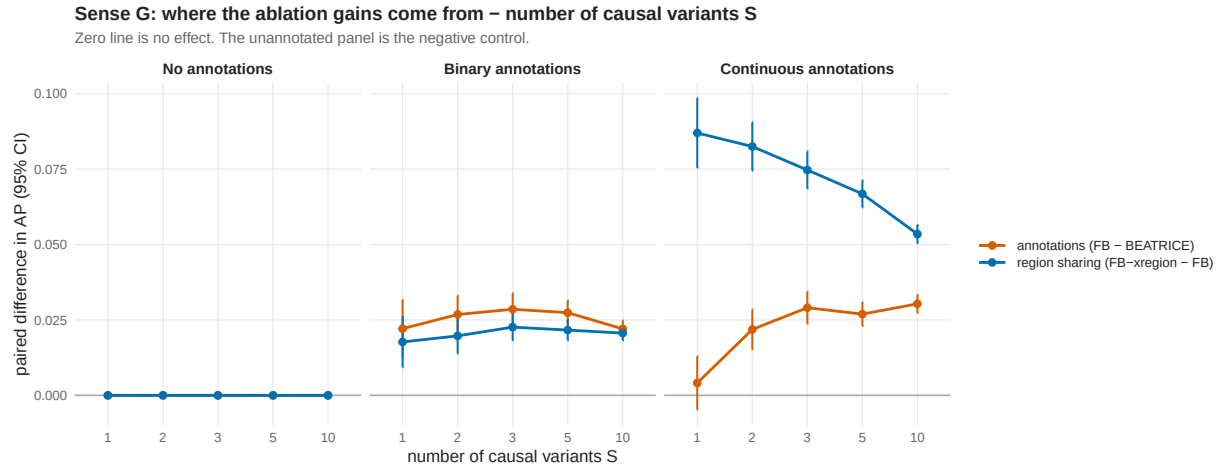


Figure 17: The two contrasts against the number of causal variants.

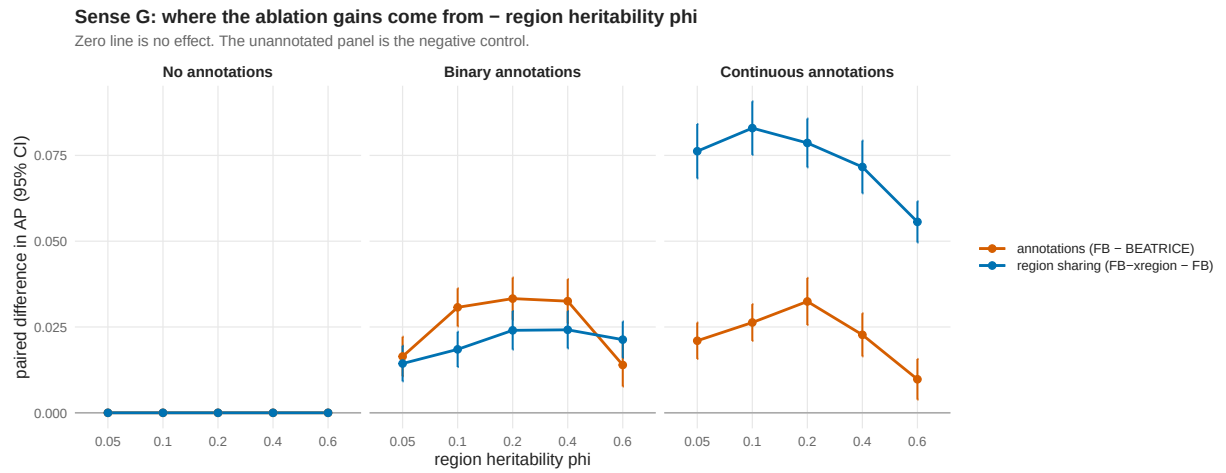


Figure 18: The two contrasts against region heritability.

Annotations alone behave oppositely at the sparse end, rising from +0.013 at $S = 1$ to a plateau around +0.027 for $S \geq 3$. With one causal variant there is little for a per-region prior network to learn from.

Against heritability both contrasts are flat across $\phi \in [0.05, 0.4]$ and dip at $\phi = 0.6$ (annotations +0.0118, sharing +0.0386) — at very high signal the likelihood dominates and the prior matters less, which is the expected Bayesian behaviour.

Against region size, sharing strengthens from +0.025 at $p = 500$ to +0.064 at $p = 1000$ and stays high, consistent with larger regions offering more variants over which a shared annotation signal can accumulate.

10.4 κ is reported but no conclusion rests on it

After gating, the residual κ estimates carry standard deviations of 1.17–1.29 against means between -0.11 and 0.31 ; `fb_pooled` reaches -0.689 on the binary arm. The denominator is small in many surviving cells and the ratio is correspondingly unstable. We report κ for completeness and rest every conclusion in this section on the controlled pairs, which are differences rather than ratios and are consequently well behaved.

The κ variance decomposition is unavailable. The replicate-level table does not carry the region index, so σ_u^2 cannot be estimated for the derived κ and Δ responses and the §2.5 correction cannot be applied. The code refuses to report an uncorrected decomposition, which would inflate whole-plot terms by $m\sigma_u^2/2$ per degree of freedom. Recovering it requires changing the pairing unit from scenario to region.

11 Validity checks

The following are verified mechanically on every run and all pass for the sparse arm:

1. In-sample LD only — zero rows with a reference-panel size.
2. Nine job rows present, matching the grid.
3. Balanced cells per method after documented exclusions — 1,125 for every analysed method.
4. No method entirely absent outside the “none” arm.
5. `polyfun_est` = `susie` on the unannotated arm.
6. `polyfun_ldsc` = `susie` on the unannotated arm.

Checks 5 and 6 are the strongest structural tests available: both methods reduce to SuSiE analytically when there are no annotations, so any discrepancy would indicate a fault in the annotation plumbing rather than in a method. They agree to numerical precision.

12 Limitations

1. **In-sample LD only.** Iteration 003 established that in-sample and reference-panel LD can *invert* method rankings. Nothing here transfers to the reference-panel setting.
2. **Sparse architecture only.** This report covers the arm with no infinitesimal background. The sparse-plus-infinitesimal arm is a separate analysis.

3. **Range dependence.** See §8; all importance statements are relative to the ranges swept.
4. **Per-cell decisions are underpowered.** See §9.
5. **κ decomposition unavailable.** The replicate-level table does not carry the region index, so σ_u^2 is unidentifiable for the derived κ and Δ responses. The corrected decomposition is skipped rather than reported uncorrected, which would inflate whole-plot terms by $m\sigma_u^2/2$ per degree of freedom.
6. **One tuning configuration.** BEATRICE-family hyperparameters come from a single Optuna study and are held fixed; competitors run at their documented defaults.

13 Full results tables

Bold rows are the BEATRICE family. *cells* counts non-missing cells out of 1,125.

Table 1: **No annotations** — all methods, sparse arm. Ranked by mean AP. *cells* is the number of non-missing cells out of 1,125; anything below that is a genuine absence, explained in §3.

method	AP	FDR@.9	mass	ECE _{hi}	CS ₉₅	cov ₉₅	RES	BSS	cells
susie_inf	0.7095	0.0062	1.14	0.0288	4.2	19.864	0.0023	0.575	125
polyfun_est	0.6819	0.0061	1.73	0.0225	1.7	19.800	0.0024	0.579	125
polyfun_ldsc	0.6819	0.0061	1.73	0.0225	1.7	19.800	0.0024	0.579	125
polyfun_oracle	0.6819	0.0061	1.73	0.0225	1.7	19.800	0.0024	0.579	125
susie	0.6819	0.0061	1.73	0.0225	1.7	19.800	0.0024	0.579	125
finemap	0.6767	0.0442	1.16	0.0683	1.5	19.168	0.0021	0.490	125
finemap_inf	0.6711	0.1815	1.82	0.1619	1.7	16.864	0.0021	0.056	120
beatrice	0.6659	0.0237	1.84	0.1434	1.9	18.888	0.0018	0.321	125
fb_pooled	0.6659	0.0237	1.84	0.1434	1.9	18.888	0.0018	0.321	125
fb_xregion	0.6659	0.0237	1.84	0.1434	1.9	18.888	0.0018	0.321	125
functional_beatrice	0.6659	0.0237	1.84	0.1434	1.9	18.888	0.0018	0.321	125
sparsepro	0.5737	0.1556	1.45	0.1232	1.5	17.336	0.0019	0.249	125
finimom	0.5735	0.1213	1.12	0.0890	1.7	18.056	0.0019	0.322	125
paintor	0.5247	0.0570	0.74	0.0615	1.8	19.008	0.0013	0.372	125
sbayesrc	0.4841	0.3705	10.02	0.2207	1.0	13.064	0.0015	-0.552	125
marginal_z	0.4086	0.0000	0.43	NaN	735.6	20.000	0.0000	-0.003	125
abf	0.4005	0.0054	0.43	0.0237	3.3	19.920	0.0008	0.314	125
funmap	NaN	0.0000	NaN	NaN	NaN	0.000	NaN	NaN	0

Table 2: **Binary annotations** — all methods, sparse arm. Ranked by mean AP. *cells* is the number of non-missing cells out of 1,125; anything below that is a genuine absence, explained in §3.

method	AP	FDR@.9	mass	ECE_{hi}	$ CS_{95} $	cov ₉₅	RES	BSS	cells
polyfun_oracle	0.8432	0.0110	1.63	0.0715	1.2	19.866	0.0032	0.694	500
polyfun_est	0.7914	0.0046	1.81	0.0376	1.4	19.914	0.0028	0.644	500
polyfun_ldsc	0.7732	0.0209	1.81	0.0440	1.3	19.668	0.0027	0.635	500
fb_xregion	0.7484	0.0569	2.08	0.1454	1.3	19.306	0.0023	0.301	500
susie_inf	0.7470	0.0043	1.16	0.0283	2.1	19.870	0.0026	0.620	500
funmap	0.7462	0.0866	1.81	0.0746	1.3	18.616	0.0026	0.614	500
functional_beatrice	0.7280	0.0783	2.10	0.1547	1.3	19.034	0.0022	0.245	500
susie	0.7232	0.0050	1.80	0.0218	1.4	19.856	0.0026	0.624	500
finemap	0.7147	0.0447	1.14	0.0648	1.3	19.346	0.0023	0.535	500
finemap_inf	0.7096	0.1717	1.76	0.1585	1.5	16.894	0.0024	0.141	480
beatrice	0.7026	0.0235	1.79	0.1592	1.9	19.050	0.0021	0.369	500
fb_pooled	0.6661	0.0745	2.09	0.1547	1.5	18.312	0.0019	0.187	500
paintor	0.6458	0.0668	0.74	0.0919	1.1	19.064	0.0016	0.452	500
finimom	0.6182	0.1156	1.09	0.0876	1.4	18.264	0.0022	0.378	500
sparsepro	0.6107	0.1509	1.45	0.1232	1.3	17.504	0.0021	0.286	500
sbayesrc	0.5086	0.3592	4.74	0.2470	1.0	13.186	0.0018	-0.313	500
marginal_z	0.4467	0.0000	0.43	NaN	765.5	20.000	0.0000	-0.003	500
abf	0.4357	0.0047	0.43	0.0255	2.7	19.914	0.0009	0.334	500

Table 3: **Continuous annotations** — all methods, sparse arm. Ranked by mean AP. *cells* is the number of non-missing cells out of 1,125; anything below that is a genuine absence, explained in §3.

method	AP	FDR@.9	mass	ECE_{hi}	$ CS_{95} $	cov ₉₅	RES	BSS	cells
polyfun_oracle	0.9114	0.0119	1.21	0.0791	1.0	19.952	0.0037	0.774	500
polyfun_est	0.8061	0.0051	1.77	0.0414	1.6	19.928	0.0028	0.641	500
fb_xregion	0.7933	0.0491	2.14	0.1529	1.3	19.462	0.0027	0.327	500
polyfun_ldsc	0.7914	0.0047	1.77	0.0360	1.6	19.906	0.0028	0.638	500
fb_pooled	0.7681	0.0422	2.15	0.1504	1.6	19.424	0.0023	0.278	500
funmap	0.7617	0.0227	1.77	0.0367	1.6	19.610	0.0027	0.628	500
susie_inf	0.7398	0.0053	1.13	0.0284	3.0	19.886	0.0026	0.618	500
functional_beatrice	0.7204	0.1002	2.27	0.1728	1.2	18.680	0.0022	0.117	500
susie	0.7175	0.0059	1.77	0.0236	1.6	19.852	0.0027	0.621	500
finemap	0.7071	0.0492	1.16	0.0682	1.4	19.250	0.0023	0.529	500
finemap_inf	0.7053	0.1708	1.77	0.1555	1.6	16.870	0.0024	0.143	480
paintor	0.7001	0.0520	0.74	0.0790	1.0	19.420	0.0019	0.520	500
beatrice	0.6981	0.0198	1.82	0.1531	1.9	18.846	0.0021	0.362	500
sbayesrc	0.6192	0.3000	5.75	0.1993	1.0	14.482	0.0024	-0.254	500
finimom	0.6177	0.1152	1.09	0.0910	1.6	18.308	0.0022	0.378	500
sparsepro	0.6093	0.1502	1.45	0.1236	1.4	17.636	0.0021	0.294	500
marginal_z	0.4371	0.0000	0.43	NaN	738.4	20.000	0.0000	-0.003	500
abf	0.4271	0.0024	0.43	0.0237	3.0	19.910	0.0009	0.333	500

Table 4: Paired ablation on AP. Each row compares two methods on the *same* simulated replicate; n is the number of paired replicates. $t = \bar{d}/\text{SE}(\bar{d})$.

stratum	contrast	$\bar{d}$	sd	n	SE	t
No annotations	annotations	0.0000	0.0000	1250	0.00000	—
No annotations	region sharing	0.0000	0.0000	1250	0.00000	—
No annotations	combined	0.0000	0.0000	1250	0.00000	—
Binary annotations	annotations	0.0254	0.0975	4981	0.00138	18.4
Binary annotations	region sharing	0.0205	0.0859	4980	0.00122	16.8
Binary annotations	combined	0.0458	0.1026	4980	0.00145	31.5
Continuous annotations	annotations	0.0224	0.0962	4965	0.00136	16.4
Continuous annotations	region sharing	0.0730	0.1183	4960	0.00168	43.5
Continuous annotations	combined	0.0954	0.1352	4960	0.00192	49.7